Gour Gupal Talukder Shawon

Shahjalal University of Science and Technology, Sylhet, Bangladesh
Sylhet, Bangladesh | gourgupaltalukder@gmail.com | +880 1746-244930
Portfolio | LinkedIn | github.com/gfor-gour | Google Scholar

Education

Shahjalal University of Science and Technology — *Sylhet, Bangladesh*
B.Sc. in ***Software Engineering*** *CGPA:* ***3.52****/4.00* — Mar 2022 - Sep 2026

Research Experience

HiveMind Research Group — *Remote*
Research Contributor — Jan 2026 - Present

Paper: *AgentCollabBench: Diagnosing When Good Agents Make Bad Collaborators* — ***NeurIPS 2026***
Resources: arXiv Preprint | GitHub | HuggingFace Dataset — *FAGEN, ICML 2026*

- Contributed to *AgentCollabBench*, a benchmark for evaluating collaboration failures and reliability in multi-agent LLM systems.
- Assisted with dataset design, benchmark construction, and evaluation setup for Software Engineering, DevOps, and Data Engineering tasks.
- Participated in experiment design, dataset validation, and research discussions within a distributed AI research team.

Paper: *Relevance is Not Obligation: Action-Relative Memory Activation for Tool-Using Agents*
Status: *Under Review (ICLR 2027)*

- Contributed to the formulation of a dynamic obligation-aware framework that evaluates memory-action pairs at the action boundary to enforce hard constraints and mitigate agent execution failures
- Spearheaded dataset annotation and quality control for coding-domain memory–action pairs, validating action scopes, footprint parameters, and enforcement tiers under near-miss interference.

Undergraduate Thesis — May 2025 – Nov 2025
Commit Message Generation using LLMs with Issue-Based Context — *Under Review (Q4)*

- **Co-First & Corresponding Author**: Led experimental design, benchmark curation, human evaluation pipeline, and paper manuscript.
- Evaluated GPT-4 Turbo, Claude 3.5, and Gemini 1.5 Pro on 309 issue-linked commits from real-world open-source repositories. [HuggingFace Dataset]
- Developed a dual evaluation framework combining automated NLP metrics and double-blind assessment by 30 software developers.
- Improved commit message quality by 42–53% over human-written baselines through prompt engineering.

Advanced Machine Intelligence Research Lab, AIUB — *Remote*
Intern Research Assistant — Feb 2026 - Jun 2026

- Contributing to research on federated unlearning for LoRA-tuned LLMs, focusing on privacy-preserving machine learning and machine unlearning.
- Assisting with literature review, experimental design, and evaluation for federated learning and parameter-efficient fine-tuning methods.
- Supporting an ongoing computer vision project on diffusion-based restoration of snow-degraded images.

Competitive Programming

- Solved **1100+** programming problems across LeetCode, Vjudge, and Codeforces with a focus on algorithms, data structures, and problem-solving.
- Maintained a **433**-day LeetCode streak, earned **23** badges, and achieved a peak rating of **1622**.

Industry Experience

ShellBeeHaken Ltd. *Dhaka, Bangladesh*
Intern Software Engineer (Onsite) Dec 2025 - May 2026

- Contributed to **production UI components** and navigation flows using **React Native (Expo)** and **Next.js**, integrating **REST APIs** and managing application state across screens.
- Assisted in building **event-driven backend services** with **AWS Lambda**; gained hands-on exposure to distributed systems concepts including **data replication** and **load balancing** during architecture discussions.
- Participated in the full **API development cycle** from writing data contracts to **integration testing**, while working in an **Agile** team with sprint planning and code reviews.

Projects

Personality Detection Using Machine Learning GitHub | Live

- Improved model **F1 score by 20%** via advanced **feature engineering** and **hyperparameter tuning**, achieving **91% accuracy** in real-time personality prediction.
- Tools: Python, scikit-learn, Flask, Next.js, Pandas, NumPy, joblib.

Local Household Service Provider Platform GitHub | Live

- Designed an **optimized data-fetching architecture** that reduced retrieval time by **60%**, resulting in **20% higher** overall application responsiveness and user satisfaction.
- Integrated **Google Gemini-powered AI chatbot**, improving service query response rate by **60%**; authored critical system documents including SRS and project proposal.
- Tools: Next.js, Express.js, MongoDB, Nodemailer, Cloudinary.

Real-Time Multi-Module Chat Application GitHub

- Built a **real-time desktop chat system** using **Java, Socket.IO (Netty), PostgreSQL, and Swing**, structured with a **multi-module Maven architecture** and shared DTOs for consistent client-server communication.
- Implemented **WebSocket-based messaging** with a custom **reconnection strategy (exponential backoff)** and heartbeat monitoring to ensure resilient real-time connectivity.
- Designed an **event-driven backend architecture** supporting authentication, messaging, and file transfer workflows, with secure and efficient persistence using **HikariCP, PostgreSQL indexing, and BCrypt authentication**.

Technical Skills

Languages	JavaScript, Java, C++, Python, TypeScript
Frontend	Next.js, React, React Native, Flutter
Backend	Node.js, Express.js, RESTful APIs, GraphQL, Socket.IO
Databases	MongoDB, Firebase, MySQL, PostgreSQL
Tools	AWS, Docker, Git, Linux, Serverless (AWS Lambda), Postman, Maven

Academic Honors & Certifications

- **2nd Runner-up** - "Bridging Engineering & Computer Science" International Hackathon, organized by ILM-AI, co-hosted by **St. George's, University of London**. Built a browser-based projectile motion simulator with optional air resistance (two-member team). GitHub | Live
- **Certifications:** AWS Cloud Practitioner · AWS Lambda · AWS DynamoDB · AWS API Gateway · Claude Code in Action (Anthropic) · Generative AI (Udacity & AWS) · ML Crash Course (Google).